Cell injury

Biochemical mechanisms:

1. ATP depletion
2. Mitochondrial damage
3. Ca2+ influx and loss of calcium homeostasis
4. Accumulation of oxygen-derived free radicals (oxidative stress)
5. Defects in membrane permeability
6. Damage to DNA and proteins (misfolded proteins)

Phase 1: The Immediate Metabolic Crisis (Ischemia/Hypoxia)

When blood flow is obstructed (**ischemia**) or oxygen supply drops (**hypoxia**), the cell experiences an immediate collapse of its energy production machinery.

- **Failure of Oxidative Phosphorylation:** The mitochondria are deprived of oxygen, the final electron acceptor in the electron transport chain. The flow of electrons halts, the proton gradient across the inner mitochondrial membrane collapses, and the synthesis of ATP drops precipitously.
- **Acceleration of Anaerobic Glycolysis:** To maintain survival, the cell shifts to emergency anaerobic pathways to generate ATP from glycogen reserves. This pathway is highly inefficient and rapidly consumes stored glucose.
- **Intracellular Acidosis:** Anaerobic glycolysis generates a massive accumulation of **lactic acid** and inorganic phosphates. Hydrogen ions (H^+) flood the cytosol and pass into the nucleoplasm, driving the intracellular pH down from its physiological baseline (~7.2) into a highly acidic range.
- **Chromatin Clumping:** The excess hydrogen ions interact directly with basic amino acid residues (lysine and arginine) on histone proteins. This protonation alters their net molecular charge and shape, disrupting the regular electrostatic repulsion between chromatin fibers. The DNA-protein complexes lose solubility, denature, and collapse inward, resulting in the **clumping of nuclear chromatin** along the nuclear periphery.

Phase 2: Reversible Injury, Pump Failure, & Hydropic Dynamics (Lecture Checklist Grounding)

As ATP concentrations plunge below critical thresholds, energy-dependent transmembrane active transport systems shut down, destroying the cell's ionic and osmotic equilibrium. During this phase, membranes and organelles are swollen and stressed, but they remain **physically intact and unbroken**.

- **Failure of the Na^+/K^+-ATPase Pump:** Lacking ATP to drive its conformational changes, this primary active transporter stops ejecting 3 Na^+ and importing 2 K^+ leaves the cell, while sodium leaks passively inward down its concentration gradient and becomes trapped.
- **Reversal of the Na^+/Ca^{2+} Exchanger (NCX):** Normally, the NCX uses the energy of sodium rushing *in* to throw calcium *out*. However, because the inside of the cell is now full of sodium, the sodium concentration gradient flattens or flips. The NCX runs in reverse: it lets trapped sodium out, but in doing so, it drags **calcium (Ca^{2+}) into the cytosol**.
- **Shutdown of Primary Calcium Pumps:** Simultaneously, the SERCA (endoplasmic reticulum) and PMCA (plasma membrane) Ca^{2+}-ATPase pumps fail due to the absolute lack of ATP, preventing the cell from sequestering calcium back into storage or pumping it out.

Histopathological & Ultrastructural Model: Cellular Swelling (Hydropic Change)

The trapped intracellular sodium drastically raises the intracellular osmotic pressure. To restore equilibrium, water rushes into the cell through aquaporins, partitioning into three distinct compartments that change the physical presentation of the tissue (e.g., hepatocytes or myocardial cells):

1. **Generalized Cytosolic Flooding (Cloudy Swelling):** Water directly expands the fluid matrix of the cell (the cytosol). This dilutes the cytoplasm, causing the entire cell to physically expand and look bloated. Under a **light microscope**, this dilution causes the cytoplasm to look pale, washed out, and "cloudy."
2. **Dilation of the ER with Ribosome Detachment (Vacuolar Degeneration):** The ER is highly sensitive to the sudden influx of water and solutes. The cisternae (normally flat tubules and sacs) of both the rough and smooth ER fill with fluid and balloon outward into massive sacs. Under a **light microscope**, these enormously distended ER compartments appear as clear, empty, water-filled bubbles or **vacuoles** within the pale cytoplasm, driving the classic term **vacuolar degeneration**. Under an **electron microscope (TEM)**, these present as markedly dilated ER tubules where the bound **ribosomes physically pop off** the rough membrane surface and scatter into the cytosol.
3. **Mild Cytoplasmic Eosinophilia (↓ Cytoplasmic RNA):** Right alongside this ER dilation and ribosome detachment, a distinct early staining shift occurs. Because ribosomes contain highly acidic RNA that normally binds the blue basic dye (Hematoxylin), the structural displacement and initial enzymatic degradation of this cytoplasmic RNA strips away the cell's baseline purple-blue tint. Consequently, the cytoplasm takes on a **mildly increased pink appearance (early eosinophilia)** purely due to the relative loss of RNA, preceding any full protein denaturation.
4. **Mitochondrial Swelling:** Water crosses the outer mitochondrial membranes into the internal matrices, causing the mitochondria to physically expand. Under the electron

microscope, small, dark, amorphous (protein-rich) densities begin to deposit within the matrix, though the inner and outer membranes remain structurally continuous.

Additional Reversible Ultrastructural Features & Inclusions

- **Plasma Membrane Blebbing and Loss of Microvilli:** As the plasma membrane distends due to internal osmotic water pressure, it undergoes specific physical distortions visible under an electron microscope:
    - *Loss of Microvilli:* Specialized surface extensions completely flatten out as the cell membrane stretches taut.
    - *Plasma Membrane Blebbing:* The membrane detaches slightly from the underlying, weakened cytoskeleton, forming outward bulges or pockets called membrane blebs.
    - *Loosening of Intercellular Junctions:* The tight connections holding neighboring cells together begin to pull apart.
- **Fatty Change (Steatosis - Especially Liver, Myocardium):** Concurrently, early cytoplasmic inclusions develop. In metabolic organs like the **liver** and **myocardium**, the lack of ATP disrupts normal fatty acid beta-oxidation and lipoprotein synthesis. Triglycerides cannot be processed or exported, leading to the accumulation of clear **lipid vacuoles** within the cytoplasm alongside the water vacuoles.
- **Contained Myelin Figures:** Early myelin figures (whorled, rolled-up masses of phospholipids resembling onions under the TEM) form from damaged, degraded organelle membranes but remain safely contained within the cytoplasm.

Phase 3: The Irreversible Turning Point (The Three Pillars of Catastrophic Structural Damage)

If the ischemia persists, the physical strain of swelling combined with the rising levels of intracellular calcium crosses a lethal threshold. The cell transitions from reversible distress to irreversible destruction, characterized by a shift from intact but distorted organelles to absolute structural breakdown:

1. Total Membrane Rupture & Structural Dissolution

- **Plasma Membrane Rupture:** The physical stretching of cellular swelling combined with the activation of calcium-dependent **phospholipases** cleaves membrane phospholipids. The plasma membrane loses its structural integrity, developing literal tears, fractures, and fragmentation.
- **The 10,000-Fold Gradient Collapse:** Normal extracellular calcium (~1–2 mmol/L) is roughly 10,000 times higher than intracellular cytosolic calcium (~0.1 μmol/L). The moment the plasma membrane develops literal tears, the structural dam collapses.

Calcium violently floods **down its steep concentration gradient, rushing from the outside directly into the cell** in a massive passive influx.

- **Lysosomal Membrane Rupture:** The fragile lipid bilayer surrounding lysosomes is broken open by the combined assault of physical swelling and baseline free radical damage.
- **Spillage of Acid Hydrolases:** Highly destructive, acidic digestive enzymes (**proteases, nucleases, cathepsins, phosphatases**) leak directly out into the cytosol.
- **Enzyme Activation via Acidosis:** Because Phase 1 created a highly acidic intracellular environment via lactic acid accumulation, **the spilled acid hydrolases are immediately activated in the cytosol**, initiating **autolysis** (self-digestion of the cell).
- **Irreversible Ultrastructural Mitochondrial Collapse:** The mitochondria transition from simple swelling to total destruction. Under the electron microscope, they exhibit **large, dense flocculent masses** (prominent lipid-protein aggregates) in the matrix, accompanied by physical disruption of the inner cristae and rupture of the outer mitochondrial membrane.
- **Uncontained Myelin Figures:** The whorled phospholipid masses grow abundant, escape containment, and begin to fill the necrotic space.

2. Protein Modification & Structural Cross-Linking

- **Protease Activation:** The massive calcium surge activates intracellular **calpain proteases**, which systematically shred the internal structural scaffolding (actin, myosin, tubulin, and vinculin), detaching the cytoskeleton from what remains of the plasma membrane.
- **Oxidative Modification:** Basal Reactive Oxygen Species (ROS) attack the amino acid side chains of functional enzymes and structural proteins. ROS specifically oxidize the sulfur-containing sulfhydryl (-SH) groups of **cysteine** and **methionine** residues.
- **Conformational Aggregation:** This oxidation forces the creation of abnormal disulfide bonds, altering the three-dimensional folding of proteins. They form tangled, insoluble **protein cross-links and aggregates**, permanently deactivating the cell's remaining metabolic machinery.

3. Catastrophic DNA Damage

- **Backbone Cleavage:** Hydroxyl radicals (OH°) cross the nuclear envelope and directly strike the sugar-phosphate backbone of the genome, causing extensive single-strand and highly lethal double-strand DNA breaks.
- **Base Oxidation (8-OHdG):** ROS oxidize nitrogenous bases within the DNA strand, converting guanine into **8-hydroxy-2'-deoxyguanosine (8-OHdG)**. During any remaining transcription or replication attempts, 8-OHdG mispairs with adenine instead of cytosine, causing severe genetic cross-linking and permanent mutations.

Phase 4: The Branching Point — Necrosis vs. Apoptosis

Once the membranes are breached, proteins are cross-linked, and the genome is damaged, the cell's remaining residual energy levels determine the pathway of its death:

Path A: Zero Residual ATP → Autolysis and Necrosis (Advanced Histopathology)

If the injury is uniform and catastrophic, leaving no energy to run programmed pathways, the cell undergoes chaotic dissolution driven entirely by the activated enzymes from Phase 3. Under a standard light microscope using H&E staining, this manifests as two classic features:

- **Intense Cytoplasmic Eosinophilia (The Full Shift to Deep Pink):** The cell progresses from the *mild* eosinophilia of Phase 2 to a brilliant, intense pink/red stain. While the cell has already completely lost its Hematoxylin (blue) binding capacity due to the total degradation of RNA by leaked **RNases**, it now undergoes a massive increase in Eosin (pink) binding. The severe intracellular acidosis and free radical attacks cause widespread **protein denaturation**. As these intracellular proteins misfold and uncoil, they physically expose thousands of hidden basic amino acid side chains (like lysine and arginine), which bind enormous amounts of the acidic pink dye (**Eosin**).
- **Homogeneous, Glassy Cytoplasm:** The cytoplasm loses its remaining granular, organized appearance and becomes completely smooth and featureless. This occurs because calcium-activated **proteases** systematically chew up the granular structural network of the cytoskeleton, leaving behind a uniform pool of denatured protein aggregates.
- **Nuclear Dissolution Timeline:** The combination of calcium-activated endonucleases and leaked lysosomal nucleases drives three distinct morphological stages of nuclear death:
    1. *Pyknosis:* The nucleus shrinks and condenses into a small, dark, dense mass of tightly packed chromatin.
    2. *Karyorrhexis:* The pyknotic nucleus physically fragments, breaking apart into pieces scattered throughout the cytosol.
    3. *Karyolysis:* The nucleases completely digest the DNA fragments. The chromatin fades entirely, and the nucleus disappears from view under the microscope.
- **Lysis:** The cell completely liquefies, its fragmented membranes dissolve, and its toxic, enzymatic internal contents spill out into the extracellular matrix. This uncontrolled release of internal cellular debris acts as a collection of **Damage-Associated Molecular Patterns (DAMPs)**, triggering a massive, local inflammatory response.

Path B: Preserved Residual ATP → The Intrinsic Apoptotic Pathway

If the cell resides at the outer boundaries of the ischemic zone (the penumbra) where minimal ATP remains, it utilizes this energy to execute a clean, ordered suicide program:

1. **p53 Tumor Suppressor Activation:** The severe double-strand DNA breaks stabilize the **p53 protein** within the nucleus. p53 halts any metabolic activity and activates the transcription of pro-apoptotic genes (like Bax and Bak).
2. **Mitochondrial Outer Membrane Permeabilization (MOMP):** Bax and Bak form oligomeric channels in the outer mitochondrial membrane, causing it to become highly permeable.
3. **Leakage of Pro-Apoptotic Factors:** The intermembrane space leaks **Cytochrome c** and **Smac/DIABLO** into the cytosol.
4. **Neutralizing the Safety Blocks: Smac/DIABLO** binds directly to **IAPs** (Inhibitors of Apoptosis Proteins). This action uncouples the IAPs from their targets, effectively removing the cell's internal safety brakes against self-destruction.
5. **Assembly of the Apoptosome:** In the presence of remaining dATP/ATP, the leaked **Cytochrome c** binds to the cytosolic receptor **Apaf-1**. Seven of these complexes bind together into a large wheel-like structure called the **Apoptosome**.
6. **Caspase Cascade Activation:** The center of the apoptosome recruits and cleaves **Procaspase-9** into active **Caspase-9** (the initiator). Active Caspase-9 then cleaves and turns on the executioner caspases, **Caspase-3 and Caspase-6/7**.
7. **Clean Dismantling:** Caspase-3 activates **Caspase-Activated DNase (CAD)** by cleaving its inhibitor. CAD enters the nucleus and cuts the DNA into regular fragments. Simultaneously, the executioner caspases neatly cleave the structural cytoskeleton, partitioning the cell into tightly sealed, membrane-bound **apoptotic bodies**.
8. **Phagocytic Clearance:** The apoptotic bodies flip the internal phospholipid **phosphatidylserine** to the outer layer of their membrane. This acts as an "eat-me" signal. Local macrophages recognize this signal and immediately engulf and digest the fragments. Because no intracellular components ever leak into the tissue space, **this entire death pathway causes zero inflammation**.

Phase 5: The Paradox of Reperfusion Injury

If medical intervention suddenly restores oxygenated blood flow (**reperfusion**) to cells that are structurally compromised but still viable, it introduces a secondary wave of accelerated destruction.

- **The Secondary Calcium Tidal Wave:** Reperfusion floods the tissue with a continuous supply of extracellular calcium-rich blood. Because the plasma membranes are already leaky and structurally weakened from the preceding ischemia, calcium pours down its gradient into the cytosol at an accelerated rate, causing a severe **calcium overload**.

- **Opening of the Mitochondrial Permeability Transition Pore (MPTP):** The overloaded mitochondria absorb this sudden wave of incoming intracellular calcium. This massive internal stress forces open the **MPTP**, a high-conductance mega-channel in the inner mitochondrial membrane. The electrical membrane potential drops to zero, permanently halting any future ATP production and causing the mitochondria to swell and burst.
- **The Explosive ROS Burst & Histological Activation:** As fresh molecular oxygen (O_2) contacts the broken, uncoupled electron transport chains of these damaged mitochondria, it is processed incorrectly. Instead of being cleanly reduced to water, the oxygen experiences an explosive **burst of Reactive Oxygen Species (ROS)**, shedding massive concentrations of superoxide (\(\text{O}_{2}^{\bullet -}\)), hydrogen peroxide (H_2O_2), and highly reactive hydroxyl radicals (\(\text{OH}^{\bullet }\)). This massive surge exponentially accelerates **lipid peroxidation**, causes widespread **protein cross-linking**, and inflicts extensive **oxidative DNA damage (8-OHdG accumulation)**, forcing any remaining viable cells rapidly into advanced cytoplasmic eosinophilia and karyolysis.
- **Neutrophil-Mediated Amplification:** The dying cells throw out severe distress signals (cytokines and DAMPs). The newly reopened blood vessels act as a highway, delivering a massive wave of **infiltrating neutrophils** to the site of injury. Once activated, these white blood cells initiate a **respiratory burst** via NADPH oxidase, releasing a secondary, external cocktail of aggressive ROS and lysosomal enzymes. This immune response creates a zone of collateral damage, killing the surrounding healthy tissue that had successfully survived the initial ischemic event.